Financial Post
re peer review

Junk Science Week: You think 'peer review' proves anything about the reliability of science?

Donna Laframboise: Journals can define peer review however they please, with no minimum standards no enforcement mechanisms

Special to Financial Post

Special to Financial Post

June 19, 2017

5:24 PM EDT

Last Updated

June 20, 2017

3:41 AM EDT

Filed under

Whether the topic is suicide prevention in Canada, bullying n Norway or climate change around the world, we're routinely assured that government policies are "evidence-based." Science itself guides our footsteps.

But what does this actually mean? In recent decades we've been encouraged to equate academic research with accuracy and reliability. A 2015 report prepared for the U.S. National Science Foundation, however, restates the obvious: A scientific finding "cannot be regarded as an empirical fact" until it has been "independently verified" by third-party researchers who've followed the same steps and achieved similar results. Findings that haven't yet been reproduced in this manner are, in scientific terms, merely tentative.

But this sort of due diligence almost never happens. What would seem to be a logical first step in establishing evidence-based policy is routinely skipped over. No systematic reproduction of research takes place before its conclusions begin shaping government policies.

When making decisions that affect human lives, bureaucrats and politicians have long substituted the judgment of people who work in academic publishing for proper scientific verification. Journals have a vetting process called "peer review." Research findings that pass peer review, so the reasoning goes, are good enough for prime time.

But peer review is a tool invented by publishers for their own purposes. It helps them sift through hundreds of manuscripts a month. It helps them identify intriguing findings that will burnish their own prestige. Since most published research is never subjected to independent verification, journals exist in a universe that threatens them with no meaningful penalties should their peer-review process be faulty.

It often is. Referees at prestigious journals have given the green light to research that was later found to be wholly fraudulent. Eugenie Samuel Reich's 2010 book, Plastic Fantastic: How the Biggest Fraud in Physics Shook the Scientific World, reports that the elite journal, Nature, published seven papers based on data faked by a hotshot young physicist named Jan Hendrik Schön.

Journals can define peer review however they please, with no minimum standards no enforcement mechanisms

The journal Science also published his work. In one instance, his paper was accepted after being examined by a single referee whose review consisted of a single paragraph. Reich writes: "Acceptance of a paper on the basis of only one review is a clear violation of Science's published guidelines; the journal promises scientists, reporters, and other authorities that research has been reviewed by at least two experts."

Conversely, Juan Miguel Campanario describes historical as well as contemporary referees who scoffed at work that later went on to win Nobel Prizes in chemistry, physics, and medicine.

If one gets what one pays for it's worth noting that referees typically work for free. They lack the time and resources to perform anything other than a metaphorical sniff test. Nothing like an audit occurs. No one examines the raw data for accuracy or the computer code for errors. Peer review doesn't guarantee that proper statistical analyses were employed, or that lab equipment was used properly.

Anyone can start an academic journal; there are an estimated 25,000 of them now. Journals are at liberty to define peer review however they please. No minimum standards apply, and no enforcement mechanisms exist.

In 2013, Irene Hames, a spokesperson for the U.K.-based Committee on Publication Ethics, told a Times Higher Education reporter she had personally encountered journal editors "who have almost boasted" about the fact that they themselves produce fake peer reviews rather than recruiting qualified referees.

Paraphrasing their remarks, she says such editors declare: "I never have a worry about finding reviewers because I just do it myself."

Climate alarmism is just the latest 'junk science' governments use to scare us

Freshii's nutritional approach is the latest victim of the junk science of Ontario's Liberal nanny state

Even when procedures are followed in good faith, the results don't inspire confidence. A full 35 years ago, back in 1982, Douglas Peters and Stephen Ceci demonstrated peer review's random, arbitrary nature by resubmitting papers to journals that had already published them.

Cosmetic changes (including the use of fictitious author and institution names) were made to 12 papers that had appeared in a dozen highly regarded psychology journals during the previous 18–32 months.

The duplication was noticed in three instances, but the remaining nine papers underwent review by two referees each. The 16 referees (89 per cent) who recommended rejection didn't cite lack of originality as a concern. Instead, the manuscripts were rejected "primarily for reasons of methodology and statistical treatment."

Only one of the nine papers was deemed worthy of seeing the light of day the second time it was examined by reviewers at the same journal.

Small wonder that Richard Smith, a former editor of the British Medical Journal, describes peer review as a roulette wheel, a lottery, and a black box. A great deal of effort has been expended, he says, trying to demonstrate that peer review improves scientific rigour. But the evidence just isn't there.

In 2011, Richard Horton, editor of The Lancet, told a U.K. parliamentary committee that "Those who make big claims for peer review need to face up to this disturbing absence of evidence."

The Internet is currently full of stories about science's reproducibility crisis. In a 2015 editorial, Horton publicly declared what many others have also been admitting: "much of the scientific literature, perhaps half, may simply be untrue." If peer review were an effective quality control mechanism, such a statement would be unthinkable.

In a world in which published research is as likely to be wrong as it is to be right, beware the claim that the latest government policy is evidence-based. Unless the research in question has been painstakingly reproduced, we have no way of knowing if it's reliable or risible.

Donna Laframboise is the author of the 2016 report, Peer Review: Why skepticism is essential.

Free this wine! Finally a plan to break down the provincial borders blocking our bottles

Email Newsletters

Get the latest news in your mailbox as it happens

FP Comment

Terence Corcoran: It's Trudeau's chance to save us from supply management. He's blowing it

Lawrence Solomon: Trudeau will learn a painful lesson — voters really dislike climate crusading

Rachel Notley's budget buys a ticket for the 'resource roller coaster' she once ridiculed

Free this wine! Finally a plan to break down the provincial borders blocking our bottles